

Yizhuo Tan

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EXPERIENCE

Research Assistant - Computer Architecture and Security (Yale/Northwestern University)	09/2023 – Present
- Quantum Systems Lab (Yale University)	09/2024 – Present

- Conduct research on quantum computing security and architecture, emphasizing how physical- and hardware-level vulnerabilities propagate to higher-level system reliability and fault-tolerant performance
- Design and evaluate quantum computer attack models targeting hardware behavior, including readout crosstalk, long-range interference, and higher-energy disturbance mechanisms
- Design oblivious QRAM protocols that protect user access patterns and privacy under adversarial models, leveraging quantum-secure and classical cryptographic primitives
- Analyze readout-crosstalk-induced errors and information leakage and quantify their impact on quantum error correction performance and logical error rates. Plan to implement defense mechanisms and mitigation strategies, including error-aware scheduling, compilation techniques, and system-level countermeasures
- Study the interaction between physical error processes and fault-tolerant protocols, bridging device-level physics with QEC code behavior and system-level abstractions
- Collaborate closely with researchers in quantum hardware, security, and architecture to translate physical insights into practical system-level defenses

Teaching Assistant – Yale University	09/2024 - 05/2025
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- Taught Introduction to Computer Engineering course in Spring 2025 at Yale
- Taught Introduction to Electronics course in Fall 2024 at Yale

Undergrad Research Assistant - Microscale Nanoscience Laboratory, USTC (Hefei, China)	09/2021 - 03/2022
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- Aimed to perform Gaussian boson sampling to demonstrate the quantum computational advantage
- Built high-quality optical paths and reflectors, achieving a coincidence rate of 0.57 through system-level optical optimization
- Helped design a method of determining coherent light using retarders and achieved optimal deployment of the retarders with improved performance

Undergrad Research Assistant - USTC (Hefei, China)	09/2021 - 12/2021
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- Designing controlled droplet suspension equipment based on standing waves physics and ultrasonic phased arrays
- Designed and fabricated planar phased array circuits to independently adjust phase and voltage of each ultrasonic sensor, realizing stable suspension and movement of multiple rows of objects
- Simulated dynamic distribution of potential field in MATLAB based on the approximate solution of GOVCOV potential
- Validated system behavior using finite-element methods and particle tracking in COMSOL
- Demonstrated stable suspension and controlled motion of multiple objects
- **Awarded:** First Prize in the 17th Physical Innovation Research Experimental Paper Competition, USTC

Teaching Assistant - USTC (Hefei, China)	09/2022 - 12/2022
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- Taught Mechanics B course in Fall 2022 at USTC

EDUCATION

PhD in Computer Science	09/2023 - Present
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- Yale University, Connecticut, United States
 - Advisor: Jakub Szefer, Yongshan Ding

- Area: Quantum Computing Security, Architecture, and Error-Aware Systems

Bachelor of Science in Physics

09/2019 – 07/2023

- University of Science and Technology of China (USTC), Hefei, China
 - Major: Physics (Atomic and Molecular Physics track)
 - Dual Major/Minor: Computer Science and Technology

SKILLS

- **Programming & Modeling** Python, C, C++, Verilog, MATLAB, COMSOL

HONORS & AWARDS

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| • Best Paper, QSec@CCS | 2025 |
| • Best Poster Award – 1 st Place, New England Hardware Security Day | 2025 |
| • Silver Award of Excellent Student Scholarship, USTC (Top 2%) | 2022, 2021 |
| • First Prize in 17th Physical Innovation Research Experimental Paper Competition, USTC | 2021 |
| • Scholarship of Yan Jici Talent Program in Physics, USTC (Top 10%) | 2021, 2020 |
| • Second Prize in 6th China Undergraduate Physics Experiment Competition | 2020 |
| • Chen Linyi Scholarship, USTC (Top 1%) | 2020 |
| • Provincial First Prize in 35th Chinese Physics Olympiad | 2018 |

UPCOMING & PAST TALKS

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| • HASP@58th IEEE/ACM International Symposium on Microarchitecture (MICRO) , Seoul, Korea | 2025 |
| • ACM QSec@ACM Conference on Computer and Communications Security (CCS) , Taipei, Taiwan | 2025 |
| • Joint March Meeting and April Meeting - Global Physics Summit , Anaheim, CA | 2025 |
| • IEEE Workshop on Quantum Intelligence, Learning & Security (QUILLS) , Washington, DC | 2024 |
| • ACM/IEEE International Conference on Computer-Aided Design (ICCAD) , Newark, NJ | 2024 |
| • 2nd Quantum Computer Cybersecurity Symposium , New Haven, CT | 2024 |

PEER-REVIEWED PUBLICATIONS

- **Yizhuo Tan**, Jakub Szefer. “I Know What You Are Reading: Evaluating Readout Crosstalk in Cloud-based Quantum Computers” in Proceedings of ACM QSec: Quantum Security and Privacy Workshop @ ACM Conference on Computer and Communications Security, **QSec@CCS**, 2025.
- **Yizhuo Tan**, Hrvoje Kukina, Jakub Szefer. “Securing HHL Quantum Algorithm against Quantum Computer Attacks” in Proceedings of Hardware and Architectural Support for Security and Privacy @ IEEE/ACM International Symposium on Microarchitecture, **HASP@MICRO**, 2025.
- Pranet Sharma, **Yizhuo Tan**, Konstantinos-Nikolaos Papadopoulos, Jakub Szefer. “Evaluation of Noise and Crosstalk in Neutral Atom Quantum Computers” in Proceedings of IEEE International Conference on Quantum Computing and Engineering, **QCE**, 2025.
- **Yizhuo Tan**, Hrvoje Kukina, Jakub Szefer. “Study of Attacks on the HHL Quantum Algorithm” in Proceedings of IEEE Workshop on Quantum Intelligence, Learning & Security @ IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications, **QUILLS@TPS-ISA**, 2024.
- **Yizhuo Tan**, Chuanqi Xu, Jakub Szefer. “Exploration of Timing and Higher-Energy Attacks on Quantum Random Access Memory” in Proceedings of the International Conference on Computer-Aided Design, **ICCAD**, 2024.

PAPERS IN PRAPERATION

- **Yizhuo Tan**, Navnil Choudhury, Kanad Basu, Jakub Szefer. “QubitHammer: Remotely Inducing Qubit State Change on Superconducting Quantum Computers”
- Navnil Choudhury, **Yizhuo Tan**, Jiaqi Yu, Kanad Basu, Jakub Szefer. “EPAR: Electromagnetic Pathways to Architectural Reliability in Quantum Processors”
- Theodoros Trochatos, Sanjay Deshpande, Chuanqi Xu, **Yizhuo Tan**, Jakub Szefer. “Security of Quantum Computing Systems -- Proposal for Foundations and Trends in Privacy and Security Tutorial Paper”